EFY Note
The source code of this project is included in this month's EFY DVD and is also available for free download at *source.efymag.com*

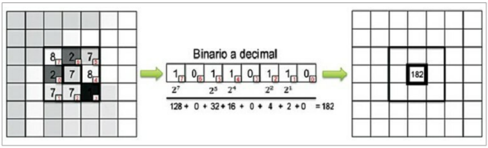

Fig. 5: Screenshot of a LBPH face recogniser

required for OpenCV and also install OpenCV 2.4.10.

```
$ sh ./install-opencv.sh
```

After installing OpenCV, check it in the terminal using import command, as shown in Fig. 2.

1. Create the database and run the recogniser script, as given below (also shown in Fig. 3). Make at least two data sets in the database.

```
$ python create_database.py person_name
```

2. Run the recogniser script, as given below:

```
$ python face_rec.py
```

This will start the training, and the camera will open up, as shown in Fig. 4. Accuracy depends on the number of data sets as well as the quality and lighting conditions.

***OpenCV* 2.4.10.** OpenCV provides the following three face recognisers:

1. Eigenface recogniser
2. Fisherface recogniser
3. LBPH face recogniser

In this project, LBPH face recognition is used, which is createLBPHFaceRecognizer( ) function.

LBP works on gray-scale images. For every pixel in a gray-scale image, a neighbourhood is selected around the current pixel and LBP value is calculated for the pixel using the neighbourhood.

After calculating LBP value of the current pixel, the corresponding pixel location is updated in the LBP mask (it is of same height and width as input image.) with LBP value calculated, as shown in Fig. 5.

In the image, there are eight neighbouring pixels. If the current pixel value is greater than or equal to the neighbouring pixel value, the corresponding bit in the binary array is set to 1. But if the current pixel value is less than the neighbouring pixel value, the corresponding bit in the binary array is set to 0. EFY

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